

DEPARTMENT OF DATA SCIENCE

INDUSTRY ORIENTED MINI PROJECT ON

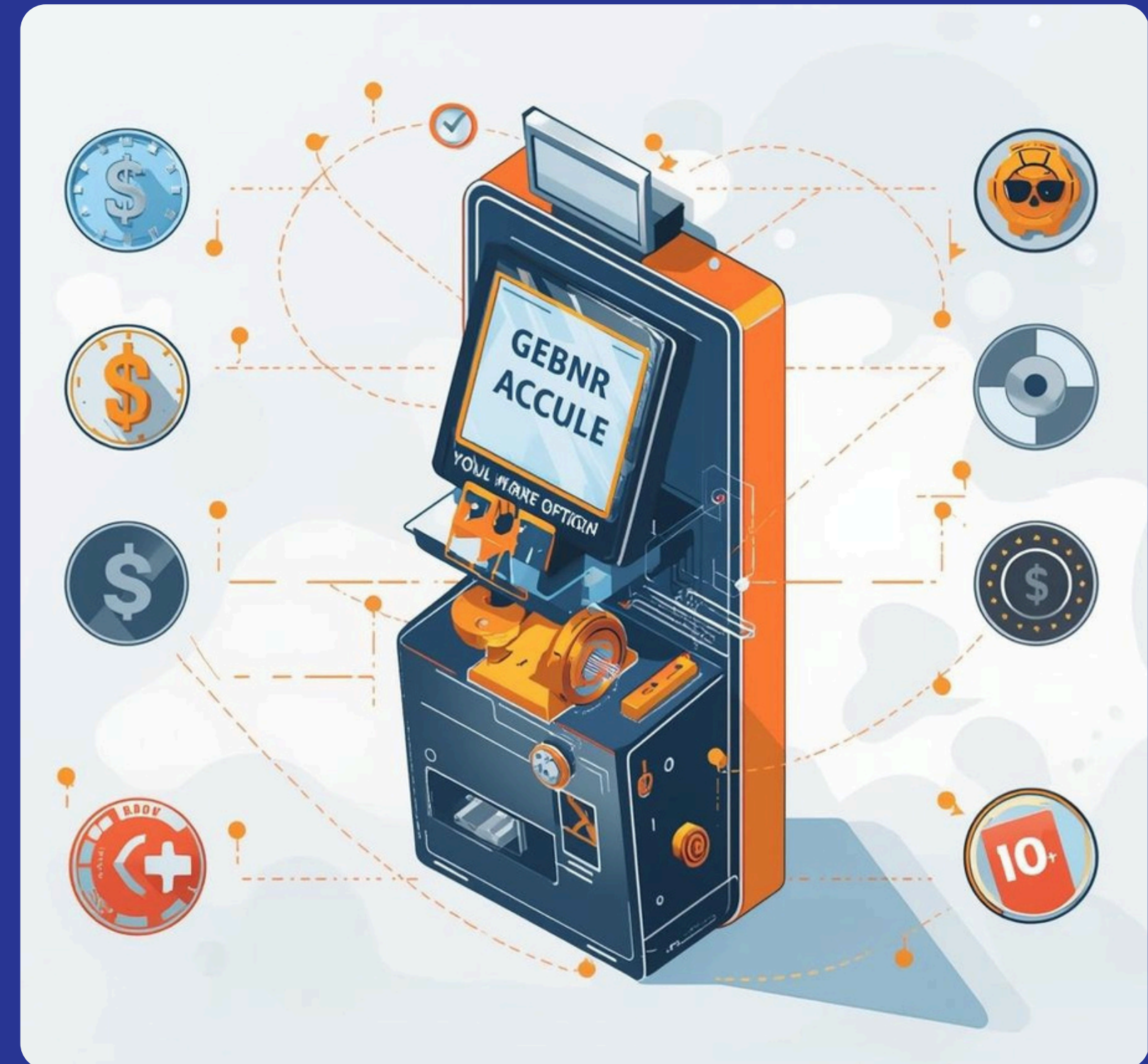
**COIN BASED MOBILE
CHARGER**

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ABSTRACT

This paper presents the design and implementation of a smart IoT-based coin-operated mobile charging system for use in public places such as railway stations, bus stands, shopping complexes, and retail outlets. The system allows users to charge mobile phones for a predefined time based on the denomination of inserted coins. A coin recognition module validates ₹1, ₹2, ₹5, and ₹10 coins and communicates with a microcontroller to assign charging durations of 5, 10, 30, and 60 minutes respectively. The microcontroller regulates a 5V power supply and controls the charging process. A reverse countdown timer displayed on an LCD shows the remaining charging time. Additional coins can be inserted during charging to extend the duration without interruption. The system features adaptive charging control to safely increase charging current for completely discharged phones. Normal charging conditions are restored once battery stability is achieved. IoT integration enables remote monitoring and usage analysis. The proposed system is cost-effective, energy-efficient, secure, and user-friendly.



INTRODUCTION

- Mobile phones are essential, but battery drain is common in public places.
- A coin-based mobile charging system provides a convenient solution.
- Users can charge their phones for a fixed time by inserting valid coins.
- The system uses a microcontroller and coin recognition module.
- It supplies a regulated 5V output for safe charging and displays time on an LCD.
- Advanced features include adaptive charging and intelligent multi-coin time allocation.



LITERATURE SURVEY

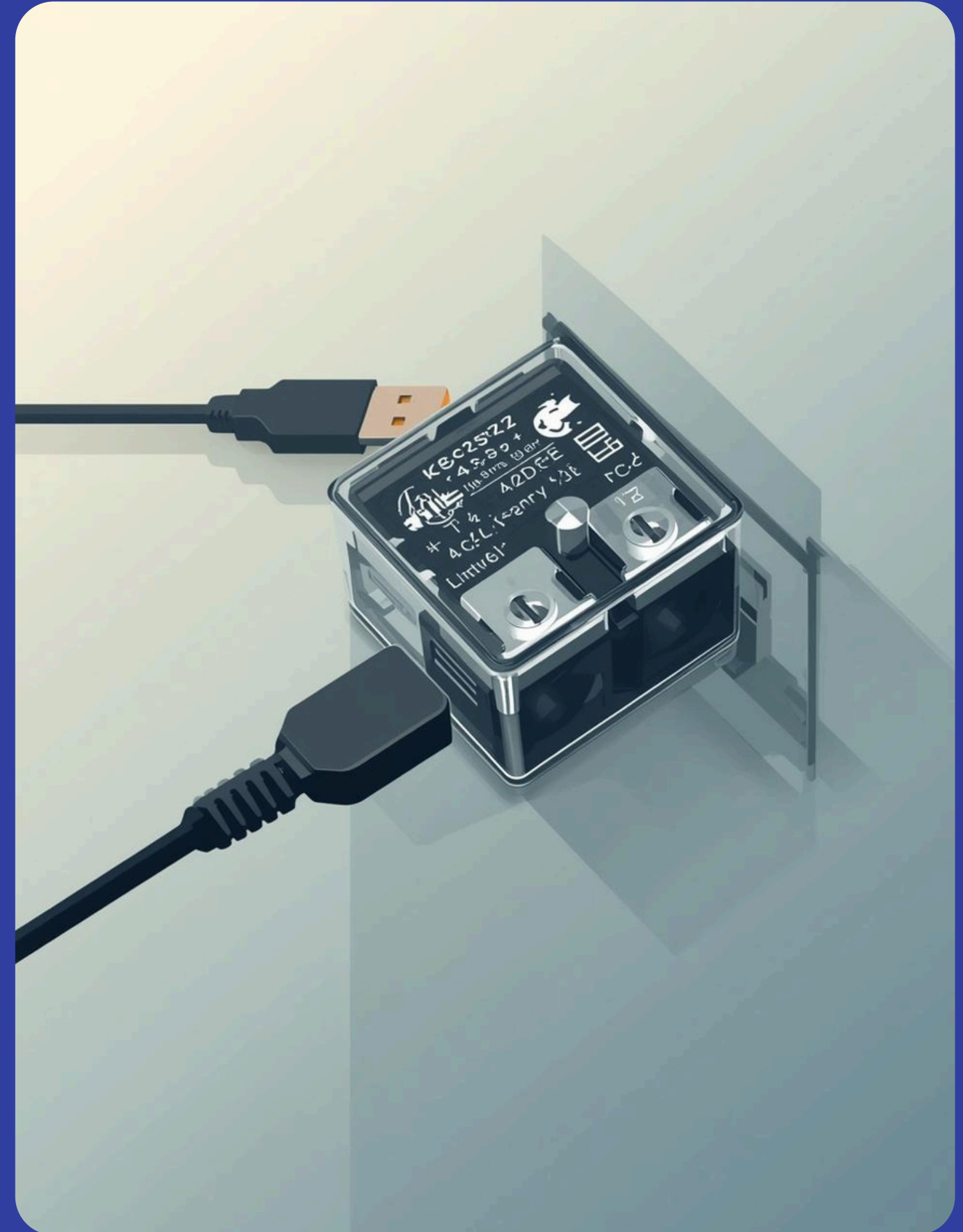
S. no	Reference / Year	Authors	Components	Key Contributions	Limitations / Observations
1.	Coin Based Mobile Charger with LCD Display (2020)	P. Ramesh et al.	Arduino UNO, Coin Sensor, LCD Display, Power Management Circuit	Added user-friendly LCD display to show charging time and status.	Limited to single-user charging at a time.
2.	Smart Coin Based Mobile Charging System (2021)	K. Lokesh et al.	Arduino, Coin Acceptor, Current Sensor, Relay Module	Introduced current sensing to prevent overcurrent and enhance safety.	No data logging or cloud connectivity.
3.	Coin Based Mobile Charger with IoT Monitoring (2022)	Md. Jaffer Sadiq et al.	Arduino, Coin Acceptor, Wi-Fi Module (ESP8266), LCD	Implemented IoT-based monitoring to track coin count and usage remotely.	Dependent on stable internet connectivity.
4.	IoT Enabled Coin Operated Mobile Charger (2023)	R. Naveen et al.	Arduino, Coin Acceptor, ESP32, Current & Voltage Sensors.	Enabled real-time monitoring of charging parameters via IoT dashboard.	Higher system cost and power consumption.
5.	Advanced Coin Based Mobile Charging Station (2024)	Komala N. et al.	Arduino MEGA, Coin Acceptor, Sensors, IoT Platform	Reviewed and implemented improvements in security, monitoring, and scalability.	Mostly prototype-based; lacks large-scale testing.

EXISTING SYSTEM

- coin-operated mobile chargers exist mainly as prototypes
- Supports only single coin denomination
- Fixed charging time per coin
- Timer resets on new coin insertion
- Charging restarts when extra coins are added
- LCD shows time only (no smart control)
- Basic IoT in some systems, not fully integrated

DISADVANTAGES:

- Not user-friendly: Remaining time gets lost
- Money wastage: Previous coin time is discarded
- Power inefficient: Frequent restart wastes energy
- No adaptive charging: Same speed for all battery levels
- Limited flexibility: No dynamic time extension
- Poor monitoring: Little or no IoT analytics

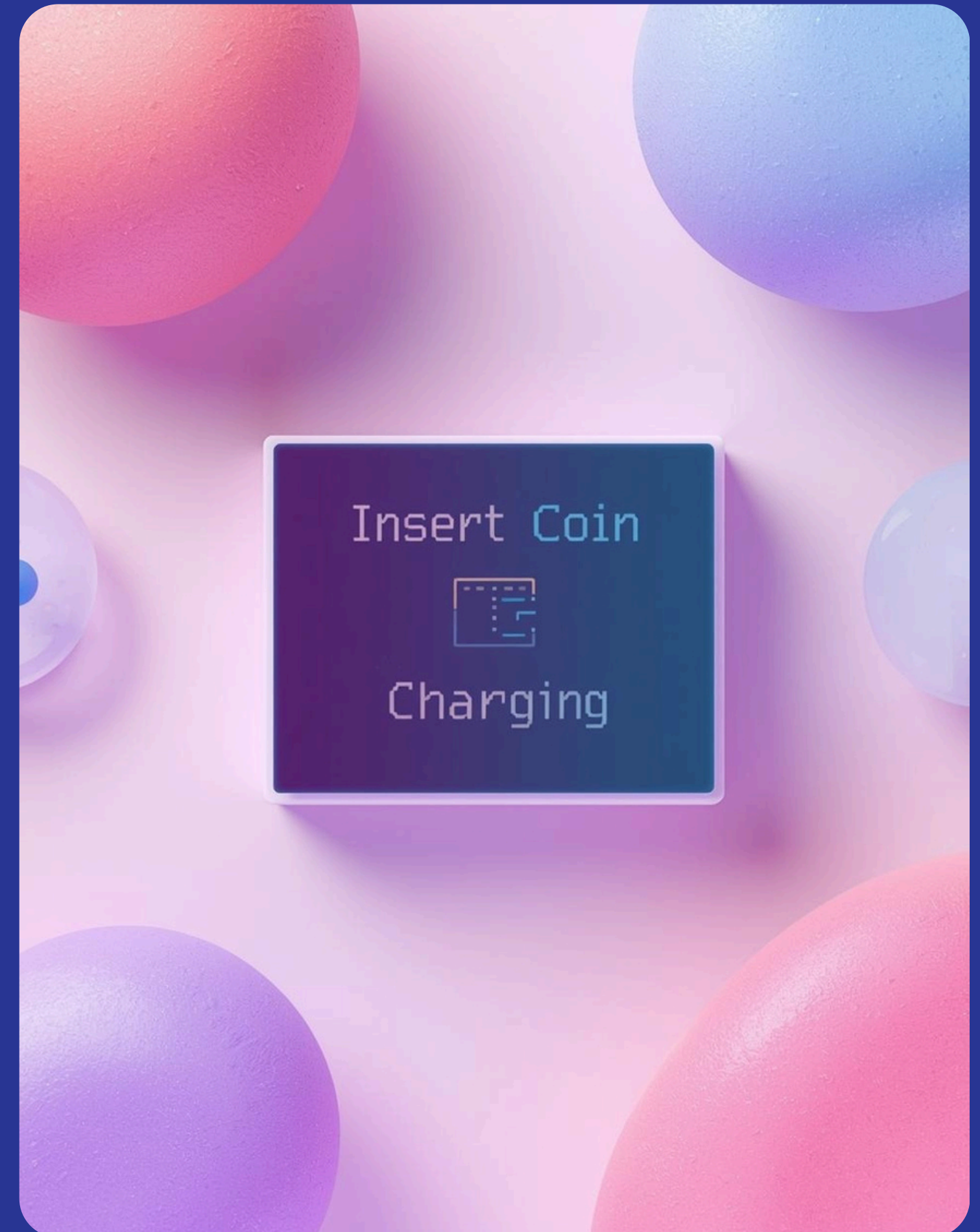


PROPOSED SYSTEM

- Accepts multiple coin denominations
- Recognizes valid coins automatically
- Allocates charging time based on coin value
- Supplies safe 5V DC for charging Displays
- remaining time on LCD Adds time without
- stopping charging Supports continuous
- charging Adapts charging based on battery
- state Enables IoT-based monitoring
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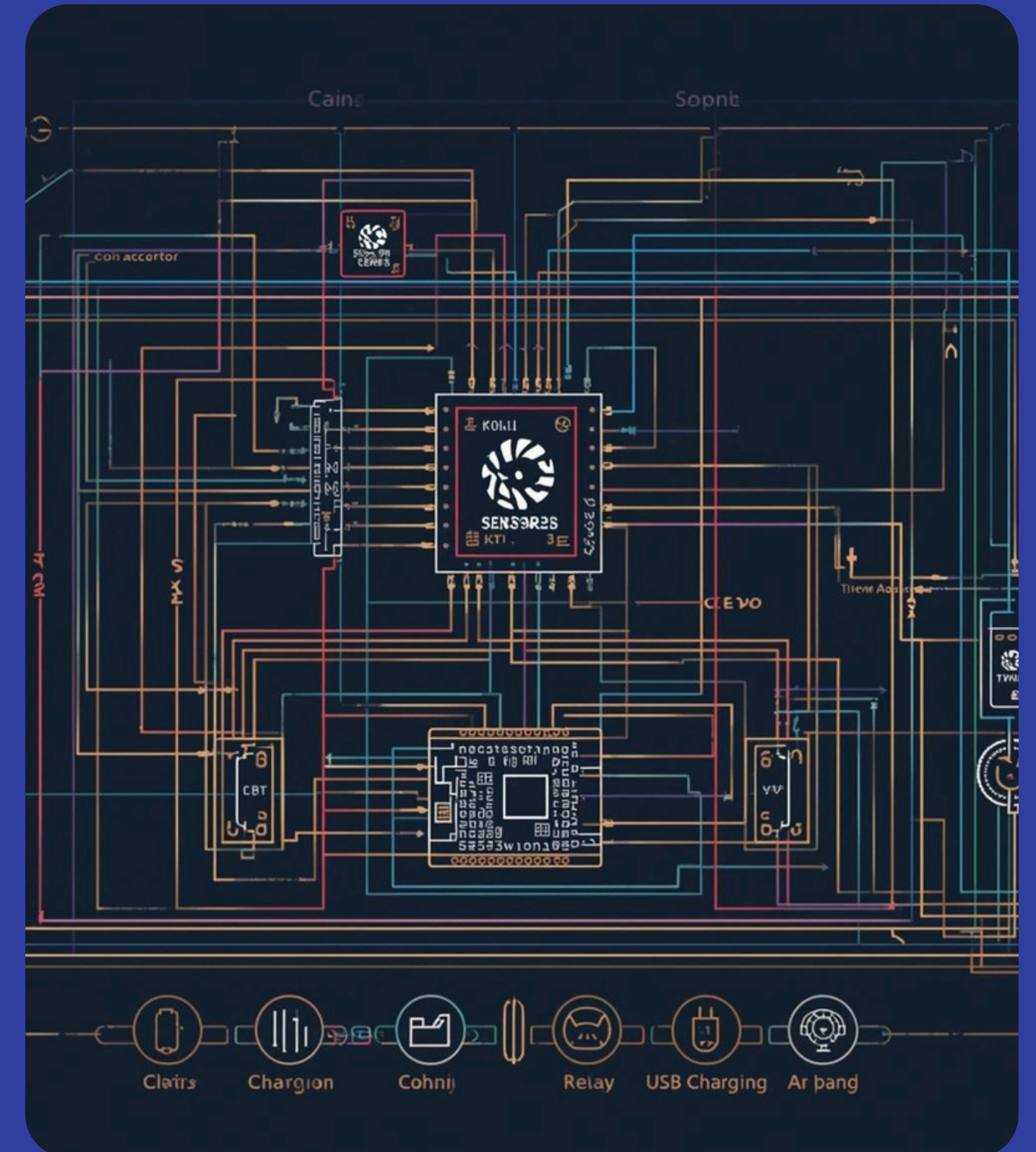
ADVANTAGES:

- No money wastage No charging restart
- User-friendly operation Smarter time
- management Energy efficient Safe
- low-voltage charging Faster initial
- charging for dead phones Suitable for
- public places Cost-effective system
- Scalable for future upgrades
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HARDWARE REQUIREMENTS

- ESP32 microcontroller (or Arduino Uno + ESP8266)
- Multi-coin acceptor module (₹1, ₹2, ₹5, ₹10)
- 12V DC power supply adapter
- Buck converter (LM2596) – 12V to 5V
- USB charging module (5V, 2A)
- Relay module or MOSFET switching module
- 16×2 LCD with I2C / OLED display
- Push buttons (reset / cancel)
- LEDs (status indicators)
- Buzzer
- PCB or breadboard
- Connecting wires and resistors
- Enclosure / casing



SOFTWARE REQUIREMENTS

1. ArduinoIDE
2. Embedded C / C++ programming language
3. ESP32 / Arduino board libraries
4. Coin acceptor pulse detection logic
5. Timer and interrupt handling code
6. Relay / MOSFET control program
7. LCD / OLED display libraries
8. Wi-Fi communication libraries (ESP32)
9. IoT cloud platform (Blynk / ThingSpeak / Firebase / MQTT)
10. Mobile app or web dashboard (optional)



Thank you!